OPIM 5509 · MODULE 2.1 · DENSE NEURAL NETWORKS

The P·E·D·W Worksheet

Forward + backprop by hand, one row of golf data - YOUR TURN

THE SETUP - ONE ROW OF THE GOLF LOGBOOK

Day D1 said: hot, humid, calm ...and the players stayed home.

Recoded for the network (1 = hot / humid / windy / played, 0 = cool / dry / calm / stayed home):

	Temp	Humidity	Wind
inputs x =	1	1	0

	Play?
target y =	0

	w1	w2	w3
weights w =	0.52	0.18	-0.12

α = 0.1 (learning rate)

The starting weights are random guesses. Your job: use P·E·D·W three times and watch them learn.

THE FOUR MOVES - SAY THEM IN ORDER, EVERY TIME

P — **PREDICTION**
$P = x \cdot w = (x1 \times w1) + (x2 \times w2) + (x3 \times w3)$ — the dot product - multiply and add

E — **ERROR**
$E = (y - P)^2$ — squared so it is always positive (just for tracking)

D — **DELTA**
$D = P - y$ — signed miss: + means we OVERestimated

W — **WEIGHT UPDATE**
new weight = old weight - α · (its input × D) — scale by the input, step against the miss

Overestimated (D > 0)? weights SHRINK. Underestimated (D < 0)? weights GROW. Input = 0? that weight does NOT move!

HOW TO USE THIS SHEET

The rectangles (vectors) are all filled in for you - you never have to hunt for a number. The formulas are yours: fill every gold-edged blank. Each round's answer feeds the next round's rectangle - so you can self-check!

ROUND 1

	Temp	Hum	Wind
x =	1	1	0

	w1	w2	w3
w in =	0.52	0.18	-0.12

P (1 × ______) + (1 × ______) + (0 × ______) = [ ]

E (0 - ______)² = [ ]

D ______ - 0 = [ ]

W w1 = ______ - 0.1×(1 × ______) ↓ w2 = ______ - 0.1×(1 × ______) ↓ w3 = ______ - 0.1×(0 × ______) ↓

w out = [ ] [ ] [ ] ← write them here, then peek at Round 2 to check!

ROUND 2

	Temp	Hum	Wind
x =	1	1	0

	w1	w2	w3
w in =	0.45	0.11	-0.12

P (1 × ______) + (1 × ______) + (0 × ______) = [ ]

E (0 - ______)² = [ ]

D ______ - 0 = [ ]

W w1 = ______ - 0.1×(1 × ______) ↓ w2 = ______ - 0.1×(1 × ______) ↓ w3 = ______ - 0.1×(0 × ______) ↓

w out = [ ] [ ] [ ] ← write them here, then peek at Round 3 to check!

ROUND 3

	Temp	Hum	Wind
x =	1	1	0

	w1	w2	w3
w in =	0.394	0.054	-0.12

P (1 × ______) + (1 × ______) + (0 × ______) = [ ]

E (0 - ______)² = [ ]

D ______ - 0 = [ ]

W w1 = ______ - 0.1×(1 × ______) ↓ w2 = ______ - 0.1×(1 × ______) ↓ w3 = ______ - 0.1×(0 × ______) ↓

w out = [ ] [ ] [ ] ← check against the strip below!

CHECK YOURSELF: P1 = 0.70 P2 = 0.56 P3 = 0.448 final w = [0.3492 , 0.0092 , -0.12]

Stuck? Say the four letters out loud - P... E... D... W - and remember: an input of 0 sends no information.